

Accessibility and usability

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Defining accessibility and usability

Accessibility

The term accessible is defined in the dictionary as “having access; easy to access or deal with, or easy to understand; intelligible”. This definition is not too far removed from the one used in the field of human-computer interaction:

“Accessibility refers to the set of features that a product, service or system must incorporate so that as many people as possible, and in as many circumstances as is commercially practical to consider, can access and use it” (Thorén, 1998).

Usability

According to the International Organization for Standardization (ISO), as stated in its standard ISO 9241-11, “Guidance on Usability” (ISO, 2018), usability is defined as the extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.

Next, we will take a closer look at each of the components of usability: effectiveness, efficiency, and satisfaction.

Effectiveness refers to the accuracy and completeness with which a user performs tasks. Here are some examples:

- A wheelchair user wants to change floors in a building. The effectiveness is the ability to move from one level to another. One solution to this would be to install a ramp.
- A screen reader user wants to read a specific section of the instruction manual for a newly purchased device. One solution is for that text to be available digitally, compatible with their screen reader.

Efficiency is the amount of resources devoted to completing a task effectively. These resources can be quantified using various measures, such as time, the number of mouse clicks, or the physical effort required. To continue with our examples:

- The wheelchair user goes up one floor using the ramp. However, if the ramp’s gradient is too steep (let’s say 17%, when building regulations state that the maximum is 11%), they will be required to exert excessive force or effort. As a result, someone will need to assist them by pushing their wheelchair, meaning their independence will be compromised. Furthermore, an excessively steep slope may cause the wheelchair to tip over, which affects both efficiency and personal safety.
- The screen reader user finds the digital version of the manual, which can be read by their screen reader, but the document is not structured into sections detectable by the screen reader. The user will be required to read the document from start to finish until they find the desired section, resulting in a waste of time.

Finally, satisfaction refers to the user’s attitude towards the system, particularly how good a person feels when using an ICT application or service. This highly subjective parameter usually concerns aspects of the user interface’s style, such as the colour scheme or the texture of the synthesised voice in a voice interface.

However, the amount of information presented to a user at any given time may also be a factor.

How to know if an ICT product is usable? This is an example of a real-life scenario involving the use of technology:

A person using email.

According to the ISO 9241-11 standard, usability refers to the extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.

This definition includes the following components:

- User type.
- Product or service.
- Specific goal (or task).
- Context.

Therefore, based on the definition of usability, the previous statement about our email user does not provide us with much information. More details about the user type, the task and the context are required. The following example shows this in action:

A blind person uses a Gmail webmail client in the Safari browser to write an email on a MacBook computer equipped with the VoiceOver screen reader.

Now there is more information available:

- The user: a blind person.
- The task: From all the possible tasks that can be carried out using email, the person is writing a message in Gmail.
- The context: They are using the Safari browser on a MacBook, which has the VoiceOver screen reader installed.

As experienced usability testers, it is clear that this person will be able to write their message efficiently with a high degree of satisfaction.

However, any changes to the parameters that define the interaction could affect this level of usability.

For example, a blind person is writing an email using the Gmail app on an Android mobile phone and the TalkBack screen reader.

The device used to perform the task changes here, and based on experience, even though both scenarios involve a blind user writing email with a screen reader, the usability can differ substantially because the interaction model, input methods, navigation structure, and accessibility maturity of the platforms are different.

ICT designers can use more information about how interaction takes place to improve their work. Let's go back to the first scenario:

A blind person uses a Gmail webmail client in the Safari browser to write an email on a MacBook computer equipped with the VoiceOver screen reader.

What additional information could be useful?

- User type: we know that the user is visually impaired. Information about the user's familiarity with screen readers or the Internet could be useful.
- The task: we know that the task is to write a message. We could include more details, such as whether blind carbon copies (BCCs) are added, whether the email is written in plain text or using an HTML editor, or whether a document needs to be attached.

- The context: This involves using the Safari browser on a MacBook, which features the VoiceOver screen reader. Further details could include the operating system and browser versions.

Link between accessibility and usability in ICT

Accessibility is often associated with how effective a task is. In other words, a product is considered accessible if the task can be performed, regardless of how much time or effort it takes.

However, in the case of a specific user type, where a product requires significantly more effort or time to use than for others, can we really consider this product to be accessible?

Additionally,

- A ramp that is too steep can pose a hazard to all users, no matter their physical abilities.
- All users benefit from a well-structured text document that allows them to use the table of contents to navigate through its sections, among other things.

Opinions vary on the link between usability and accessibility.

Some people consider them to be separate concepts. Some even work on usability with no regard for the concept of accessibility whatsoever. In essence, they focus on improving the effectiveness, efficiency and user satisfaction associated with ICT without considering functional diversity. Their focus is solely on the “standard” population if such a thing exists.

Conversely, considering functional diversity and our responsibility as designers to prevent discrimination on this basis, or even to ensure that our organisation does not breach the law, it makes sense to conclude that usability and accessibility are intrinsically linked concepts. ICT applications must therefore be both usable and accessible, so that the vast majority of people can use them effectively.

As such, no product labelled as “usable” should cause accessibility issues.

Otherwise, the product might be highly usable for some users, but its effectiveness and efficiency could drop to unacceptable levels for others, potentially preventing them from accessing certain features.

The approach adopted by the ISO 9241-171:2025 standard (ISO, 2025) is particularly relevant, as it defines accessibility as the usability of a product, service or environment by different people belonging to the widest possible range of functional abilities.

This definition matters because it gives us solutions to our dilemma:

- An application is only accessible if all user groups can use it, or at least the vast majority of people.
- A product might be usable for some user groups, but not for others. There are also ethical and legal implications to consider.

Let's think back to a product we mentioned earlier: Gmail. It is common to first define the environment (for example, the Gmail website in Safari, or the Gmail app on Android) when determining accessibility, and then decide that it is accessible if usability is sufficient:

- For all tasks supported by the product. For email, this includes writing emails, receiving alerts for new messages, identifying the sender, downloading attachments, organising messages using labels, and so on.
- For as many functional user groups as possible, i.e. for people who are visually impaired, people with low upper limb dexterity, people with cognitive disabilities, and so on.

How can we ensure that our product is accessible, meaning that it can be used by as many user groups as possible? Methodologies known as User-Centred Design have been developed to establish the phases and techniques required to achieve this end.

Firstly, all these phases must comply with the standards in which experts have agreed on a minimum set of requirements that must be met to address the needs of the majority of users. Examples include the European standard EN 301 549 on accessibility requirements for ICT products and services, or the Section 508 standards in the United States and Canada. More information on these standards will be provided later.

References

- ISO (2018). ISO 9241-11:2018 “Ergonomic requirements for office work with visual display terminals (VDTs). Part 11: Guidance on usability”. Available at: [Website with the ISO 2018 reference](#).
- ISO (2025). ISO 9241-171:2025 “Ergonomics of human-system interaction - Part 171: Guidance on software accessibility”. Available at: [Website with the ISO 2025 reference](#).
- Thorén C. (ed.) (1998). “Nordic Guidelines for Computer Accessibility”. 2nd ed., Nordic Cooperation on Disability.